

习题 1 答案

1. 根据下列瞬时值表达式写出对应复数形式。

(1) $\cos(\omega t + kz)$;

(2) $\sin kz \cos \omega t$;

(3) $e^{-\alpha z} \sin(\omega t - kz)$;

(4) $\vec{x} \cos \omega t - \vec{y} \sin \omega t$ 。

解：(1) $u(z, t) = \cos(\omega t + kz)$

其对应复数形式为： $U = e^{jkz}$

$$(2) \quad u(z, t) = \frac{1}{2} [\sin(kz + \omega t) + \sin(kz - \omega t)] = \frac{1}{2} [\cos(\omega t + kz - \frac{\rho}{2}) + \cos(\omega t - kz + \frac{\rho}{2})]$$

其对应复数形式为： $U = \frac{1}{2} [e^{j(kz - \frac{\rho}{2})} + e^{j(-kz + \frac{\rho}{2})}] = \frac{1}{2} (-je^{jkz} + je^{-jkz}) = \sin(kz)$

$$(3) \quad u(z, t) = e^{-\alpha z} \cos(\omega t - kz - \frac{\pi}{2})$$

其对应复数形式为： $U = e^{-\alpha z} e^{j(-kz - \frac{\pi}{2})} = -je^{-\alpha z - jkz}$

$$(4) \quad u(z, t) = \vec{x} \cos \omega t - \vec{y} \cos(\omega t - \frac{\pi}{2})$$

其对应复数形式为： $U = \vec{x} - \vec{y} e^{-j\frac{\pi}{2}} = \vec{x} + j\vec{y}$

2. 根据以下复数表达式写出对应瞬时值。

(1) $(1 + j)e^{-jkz}$

(2) $j \cos kz$

(3) $e^{-\alpha z} e^{jkz}$

(4) $\vec{E} = \vec{x}(2 + 3j) + \vec{y}(4 - 3j)$

解：(1) $U = (1 + j)e^{-jkz} = e^{-jkz} + je^{-jkz}$

所以

$$\begin{aligned} u(z, t) &= \operatorname{Re} \left[(e^{-jkz} + je^{-jkz}) e^{j\omega t} \right] \\ &= \cos(\omega t - kz) - \sin(\omega t - kz) \end{aligned}$$

$$(2) \quad U = j \cos kz$$

所以

$$\begin{aligned} u(z, t) &= \operatorname{Re} \left[j \cos kz e^{j\omega t} \right] \\ &= -\cos kz \sin \omega t \end{aligned}$$

$$(3) \quad U = e^{-\alpha z} e^{jkz}$$

所以

$$\begin{aligned} u(z, t) &= \operatorname{Re} \left[e^{-\alpha z} e^{jkz} e^{j\omega t} \right] \\ &= e^{-\alpha z} \cos(\omega t + kz) \end{aligned}$$

$$(4) \quad \vec{E} = \vec{x}(2 + 3j) + \vec{y}(4 - 3j)$$

所以

$$\begin{aligned} \vec{E}(z, t) &= \operatorname{Re} \left[(\vec{x}(2 + 3j) + \vec{y}(4 - 3j)) e^{j\omega t} \right] \\ &= \vec{x}(2 \cos \omega t - 3 \sin \omega t) + \vec{y}(4 \cos \omega t + 3 \sin \omega t) \end{aligned}$$